

Assignment IV

AI Compute Platforms, Applications and Trends

Submission PDF file Include:

Student Name:

Student ID:

Assignment Name:

Task 1 :Develop the Flask Application

Create a Flask web application that accepts the following information from the user:

- Student Name
- Mark

The application must determine and display the student's grade according to the following criteria:

Mark	Grade
80–100	A
70–79	B
60–69	C
50–59	D
Below 50	Fail

The application must validate the entered mark. If the mark is less than 0 or greater than 100, display an appropriate Invalid Mark message.

Test the application locally before containerizing it.

Task 2:Configure the Application Port

Configure the Flask application to:

- Listen on 0.0.0.0
- Run on port 6500

Make sure that the application can be accessed correctly when it is running inside a Docker container.

Task 3 :Create the Dockerfile

Create a Dockerfile for the Flask application.

The Dockerfile should:

- Use an appropriate Python base image.
- Copy the required application files into the image.
- Install Flask and other required dependencies.
- Configure the application to run correctly.
- Expose/document the required application port.

Task 4 :Build the Docker Image

Build a Docker image using the following image name and tag:

grade-app:v1 Verify that the image has been created successfully.

Task 5 : Create and Run the Container

Create a Docker container named:

grade1

Run the container in detached mode. Configure Docker port mapping so that:

Host Port	Container Port
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5000	6500
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The Flask application must therefore be accessible from the host machine through port 5000.

Task 6 : Verify the Container

Verify that:

- The container is running.
- The correct Docker image is being used.
- Host port 5000 is mapped to container port 6500.

Task 7 : Test the Containerized Application

Open the application in a web browser using:

http://localhost:5000

Test the application with different input values and verify that it produces the correct results.

Test Mark	Expected Result
85	Grade A
75	Grade B
65	Grade C
55	Grade D
45	Fail
120	Invalid Mark

Submission Requirements

Each student must submit **ONE PDF file** containing the following sections.

1. Student Information

Include:

Student ID:

Student Name:

Class/Batch:

Assignment Title:

2. Source Code

Include the complete source code for:

(a) Flask application

app.py

(b) Dockerfile

Dockerfile

3. Screenshots of Practical Work

Students must provide clear screenshots showing both the command and its output/result.

The PDF should contain these **7 pieces of evidence:**

No.	Required Evidence
1	Flask application successfully tested locally
2	Docker image grade-app:v1 successfully built
3	docker images showing grade-app:v1
4	Container grade1 successfully created and running
5	docker ps showing the 5000 → 6500 port mapping
6	Browser showing a successful grade result
7	Browser showing the Invalid Mark test result

For the browser result, the screenshot should clearly show enough information to verify that the application is being accessed through:

localhost:5000